SETS

Subtitle

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OUTLINE

Set

An unordered collection of entities/objects (called set members/elements of a set).

- Many discrete structures we are going to study are built using sets.Examples include:
  - Combinations- unordered collection of objects-used in counting.
  - Relations- sets of ordered pairs that represent relationships,
  - Graphs- sets of vertices and edges that connect vertices
  - Finite state machines- used to model computing machines
- Sets group objects with similar properties together. e.g. set of CSI131 students; set of odd numbers, etc.
- The order of elements not significant, repeating elements also insignificant.

Set membership

Elements placed in curly brackets: $A = \{2, 4, 6, 8\}$.

We say 2 is an element of A, i.e. $2 \in A$.

3 ways to specify sets

1. List all members of the set (**roster method**).
   - $V = \{a, e, i, o, u\}$
2. Sometimes the roster method is used without listing all the members of the set. The ellipses (...) are used when the general pattern of the elements is obvious.
   - $S = \{2, 4, 6, 8, ...\}$ - set of positive even numbers.
3. Give a property that characterizes the elements (**set builder**).
   - $B = \{x/x$ is a 2 digit integer each of whose digit is 1 or 2 $\}$.
     $B = \{11, 12, 22, 21\}$
   - $C = \{x/x$ is an odd integer $> 1\}$. $C = \{3, 5, 7, 9, ...\}$

Notation:

$x \in A$ means x is a member of set A, whereas $x \notin A$ means x is not a member of A.

1. Example: Let $A = \{1, 2, 3, 4, 5, 6\}$ and $D = \{1, 2, 3\}$ then D can be specified as:
   - $D = \{x | x \in A \wedge x < 4\}$ or $D = \{x \in A | x < 4\}$
   - In words, D is a set of all x in A such that $x < 4$
2. Example: Let $B = \{x | x$ is an odd number $> 1.\}$ and Let $C = \{3, 5, 7, 9, 11, 13\}$
   - Then $C = \{x | x \in B \wedge x \leq 13\}$ or $C = \{x \in B \wedge x \leq 13\}$

Subsets
Let A and B be sets.
A is a subset of B, written as $A \subseteq B$, if all elements of A are in B.
B is a superset of A.
$A \subseteq B$ if and only if the quantification $\forall x(x \in A \to x \in B)$ is true.

Showing that A is a Subset of B To show that $A \subseteq B$ show that if x belongs to A then x also belongs to B.
Showing that A is Not a Subset of B To show that $A \nsubseteq B$, find a single $x \in A$ such that $x \notin B$.

Equality of sets

We write $A = B$ (A and B are equal sets) if sets A and B have similar elements.
Therefore, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$

To prove that $A = B$, show that $A \subseteq B$ and $B \subseteq A$.

Set Complement

Let A be a set.

- The **complement** of set A, written as $\overline{A}$ is the set of everything that is not an element of A.
- The complement in defined in the context of some universal set U containing all elements of interest.
- $\overline{A} = \{x | x \in U \wedge x \notin A\}$.

The Empty Set

The **empty set** or **null set** is denoted by $\emptyset$ or $\{ \}$

Theorem
For every set S,
(i) $\emptyset \subseteq S$, (ii) $S \subseteq S$

The Size of the Set

Def

Let S be a set

If there are exactly n distinct elements in S, $n \geq 0$, $n \in \mathbb{Z}$, we say that S is a finite set and that n is the **cardinality** of S, denoted $\mid S \mid$.

e.g. Let A be the set of odd positive integers less than 10. Then $\mid A \mid = 5$.

e.g. The null set has no elements, it follows that $\mid \emptyset \mid = 0$. A set is said to be infinite if it is not finite.

Power Sets

Def

Given a set S, the power set of S is the set of all subsets of the set S. The power set of S is denoted by $\mathcal{P}(S)$.

If a set has n elements, then its power set has 2^n elements.

e.g What is the power set of the set {0, 1, 2}?

Sol

The power set $\mathcal{P}(\{0,1,2\})$ is the set of all subsets of {0, 1, 2}.
$\mathcal{P}(\{0,1,2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0,1\}, \{0,2\}, \{1,2\}, \{0,1,2\}\}$

Cartesian Product

Def

Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$.
Hence,

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

Sol

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$.
The Cartesian product $A \times B$ is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2.c)\}$$

Note: $A \times B$ and $B \times A$ are not equal.

Using Set Notation with Quantifiers

What do the statements $\forall x \in \mathbb{R}(x^2 \geq 0)$ and $\exists x \in \mathbb{Z}(x^2 = 1)$ mean?

$\forall x \in \mathbb{R}(x^2 \geq 0)$ can be expressed as "The square of every real number is nonnegative." This is a true statement.
The statement $\exists x \in \mathbb{Z}(x^2 = 1)$ can be expressed as "There is an integer whose square is 1." This is also a true statement because $x = 1$ is such an integer (as is -1).

Set Operations

Let A, B be sets. Set A and B can be combined using set operations.
Union - $\cup$
$A \cup B = \{x|x \in A \vee x \in B\}$. e.g: $\{8,7,3,2\} \cup \{4,8,2\} = \{2,3,4,7,8\}$

Intersection - $\cap$
$A \cap B = \{x|x \in A \wedge x \in B\}$. e.g: $\{8,7,3,2\} \cap \{2,8\} = \{2,3,4,7,8\}$

Set difference
$A - B = \{x|x \in A \wedge x \notin B\}$ which is the same as $A \cap \overline{B}$,
Example: $\{8,7,3,2\} - \{4,8,2\} = \{3,7\}$

Principle of inclusion-exclusion

- $\mid A \mid + \mid B \mid$ - counts all elements in A, then add that count to that of all elements in B. so an element that appears in both A and B is counted twice.
- So, to count elements in $A \cup B$ use the following:
  $\mid A \cup B \mid = (\mid A \mid + \mid B \mid) - \mid A \cap B \mid$

The Symmetric difference - $\oplus$
is the set containing those elements in either A or B, but not in both A and B.
$A \oplus B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

Summary

- $|A|+|B|$ - counts all elements in A, then add that count to that of all elements in B. so an element that appears in both A and B is counted twice.
- So, to count elements in $A \cup B$ use the following:
  $|A \cup B| = (|A| + |B|) - |A \cap B|$

Set Identities

Note the similarities with logical equivalences.

Set Identities.

Identity	*Name*
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws

Identity	*Name*
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Methods of Proving Set Identities

Methods of Proving Set Identities.	
Description	***Method***
Subset method	Show that each side of the identity is a subset of the other side.
Membership table	For each possible combination of the atomic sets, show that an element in exactly these atomic sets must either belong to both sides or belong to neither side
Apply existing identities	Start with one side, transform it into the other side using a sequence of steps by applying an established identity.

If **A** and **B** are sets, then,

$$A \times B = \{(x, y) : x \in A, y \in B\}$$

$$A \cup B = \{x : x \in A \wedge x \in B\}$$

$$A \cap B = \{x : x \in A \vee x \in B\}$$

$$A - B = \{x : x \in A \wedge x \notin B\}$$

$$\overline{A} = \mathrm{U} - A$$

Proof using the Subset Method

Example Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution We will prove that the two sets $\overline{A \cap B}$ and $\overline{A} \cup \overline{B}$ are equal by showing that each set is a subset of the other, i.e. $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$ and $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$
First, show that $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$, i.e. if x is in $\overline{A \cap B}$, then it must must also be in $\overline{A} \cup \overline{B}$.

- Suppose that $x \in \overline{A \cap B}$.
- By def of a complement $x \notin A \cap B$.
- By def of intersection $\neg((x \in A) \wedge (x \in B))$.
- Applying De Morgan's Law $\neg(x \in A)$ or $\neg(x \in B)$.
- By definition of negation $x \notin A$ or $x \notin B$
- By definition of complement $x \in \overline{A}$ or $x \in \overline{B}$.
- $x \in \overline{A} \cup \overline{B}$

Hence shown.

Showing that $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$
We do this by showing that if $x \in \overline{A} \cup \overline{B}$, then it must be in $\overline{A \cap B}$.

- Suppose that $x \in \overline{A} \cup \overline{B}$.
- By definition of union $x \in \overline{A}$ or $x \in \overline{B}$
- By def of a complement $x \notin A$ or $x \notin B$.
- Consequently $\neg((x \in A) \vee (x \in B))$ is true.
- Applying De Morgan's Law $\neg(x \in A)$ and $\neg(x \in B)$ is true.
- By definition of intersection $\neg(x \in (A \cap B))$ is true.
- By definition of complement $x \in \overline{A \cap B}$.

Hence shown.
$\therefore \overline{A \cap B} = \overline{A} \cup \overline{B}$

Proof using the set builder notation

Example Use set builder notation and logical equivalences to establish the first De Morgan law $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

We can prove this identity with the following steps.

$$
\begin{aligned}
\overline{A \cap B} &= \{x | x \notin A \cap B\} && \text{by definition of complement} \\
&= \{x | \neg(x \in (A \cap B))\} && \text{by definition of does not belong symbol} \\
&= \{x | \neg(x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
&= \{x | \neg(x \in A) \vee \neg(x \in B)\} && \text{by the first De Morgan law for logical equivalences} \\
&= \{x | x \notin A \vee x \notin B\} && \text{by definition of does not belong symbol} \\
&= \{x | x \in \overline{A} \vee x \in \overline{B}\} && \text{by definition of complement} \\
&= \{x | x \in \overline{A} \cup \overline{B}\} && \text{by definition of union} \\
&= \overline{A} \cup \overline{B} && \text{by meaning of set builder notation}
\end{aligned}
$$

Hence shown.

Proof using Membership Table

Example Use a membership table to show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

A Membership Table for the Distributive Property.							
A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Example 1 Prove that for all sets A and B, $A \cap B \subseteq A$

Proof: Suppose A and B are any sets.
Must show: $A \cap B \subseteq A$
Let $x \in A \cap B$
Then, $x \in A \land x \in B$, definition of Intersection
In particular, $x \in A$
Thus, every element in $A \cap B$ is in A.
$\therefore A \cap B \subseteq A$

Example 2 Prove that for all sets A and B, $A \subseteq A \cup B$

Proof: Suppose A and B are any sets.
Must show: $A \subseteq A \cup B$
Let $x \in A$. Then, the following statement is true,
$x \in A$ or $x \in B$, definition of Union
In particular, $x \in A \cup B$
Thus, every element in A is in $A \cup B$.
$\therefore A \subseteq A \cup B$

Example 3 Prove that for a set A, $A \cap A \subseteq A$

Proof: Suppose A is any set.
Must show: $A \cap A \subseteq A$
Let $x \in A \cap A$. Then, the following statement is true,
$x \in A$ and $x \in A$, definition of Intersection
Thus, $x \in A$
That is, every element of $A \cap A$ is in A.
$\therefore A \cap A \subseteq A$

Example 4 Prove that for a set A, $A \cup (A \cap B) \subseteq A$

Proof: Suppose A and B are any sets.
Must show: $A \cup (A \cap B) \subseteq A$
Let $x \in A \cup (A \cap B)$. Then, the following statement is true,
$x \in A$ or $x \in (A \cap B)$, by Definition of Union
Break into cases:
Case 1: $x \in A$, then $x \in A$.
Case 2: $x \in A \cap B$,
$x \in A$ and $x \in B$, So $x \in A$.
Thus, in both cases, $x \in A$
That is, every element of $A \cup (A \cap B)$ belongs to A.
Hence, $A \cup (A \cap B) \subseteq A$

Example 5 Prove that for a set A, $A \cup (A \cap B) = A$

Proof: Suppose A and B are any sets.
Must show: $A \cup (A \cap B) \subseteq A$ and $A \subseteq A \cup (A \cap B)$
First, showing that let $x \in A \cup (A \cap B)$ see proof in **Example 4.**
Second, showing $A \subseteq A \cup (A \cap B)$, Let $x \in A$.
By Definition of Union, $x \in A$ implies or $x \in (A \cup S)$ for any set S, and in particular $x \in A \cup (A \cap B)$.
Thus, $A \subseteq A \cup (A \cap B)$.
That is, every element of $A \cup (A \cap B)$ belongs to A.
Hence, $A \cup (A \cap B) \subseteq A$
$\therefore A \cup (A \cap B) = A$

Example 6 Prove that for a set A, $\emptyset - A = \emptyset$

Proof: Suppose A is any set.
Must show: $\emptyset - A = \emptyset$ and $\emptyset \subseteq \emptyset - A$
Let $x \in \emptyset - A$, then $x \in \emptyset$, which is impossible. So $\emptyset - A$ has no elements. Thus, $\emptyset - A = \emptyset$. (Or use the Definition of Set Difference)
Conversely, if $x \in \emptyset$, then $x \in \emptyset$ and $x \notin A$. Both sets are empty. Thus, $\emptyset \subseteq \emptyset - A$

Example 7 Prove that for any sets A and B, if $A \subseteq B$, then

(a) $A \cup B = A$, and (b) $A \cup B = B$

Proof: Suppose A and B are any sets.
Must show: Given $A \subseteq B$,

(a) $A \cap B \subseteq A$ and $A \subseteq A \cap B$, and (b) $A \cup B \subseteq B$ and $B \subseteq A \cup B$

(a) For $A \cap B \subseteq A$
This is always true by definition of intersection: any element in $A \cap B \subseteq A$.
For $A \subseteq A \cap B$, Let $x \in A$. Since $A \subseteq B$, $x \in B$.
Thus $x \in A$ and $x \in B$, so $x \in A \cap B$.
$\therefore A \subseteq A \cap B$.
Since $A \cap B \subseteq A$ and $A \subseteq A \cap B$, we have $A \cup B = A$.

(b) For $A \cup B \subseteq B$, let $x \in A \cup B$. Then, $x \in A$ or $x \in B$.
if $x \in A$, $x \in B$ since $A \subseteq B$.
if $x \in B$, then $x \in B$.
So, in either case $x \in B$.

For $B \subseteq A \cup B$, this is always true by definition of union: any element of B is in $A \cup B$
Thus, $A \cup B \subseteq B$ and $B \subseteq A \cup B$, so $A \cup B = B$. $\therefore A \cup B = B$

The End